

system RAM. GDDR3, a creation of ATI, seen in many present graphics cards, uses a technology distinct from that of DDR3, which is used in high-end computers. The latest graphics cards using GDDR4 also have a presence in the market.

4.2.17 HDR

High Dynamic Range is a relatively new display concept, only implemented in the latest GPUs. It refers to the range of lighting in which the detail of an image is preserved. Normally, visual elements that are too dark or too bright, for example areas that are overshadowed, are not displayed with the same amount of detail as well-lit areas. Doing so saves resources since in any case the detail in these areas are not noticeable to the eye. But in HDR mode, even these details become noticeable. Rendering in HDR requires higher computational power, which has become possible with the latest graphics cards.

4.2.18 Memory Bus Width

The width of the pathway between the Graphics RAM and the core is called the Memory Bus Width. The wider the pathway, the faster the data can be processed.

4.2.19 Memory Clock

This refers to the operating frequency of the graphics RAM on the graphics card. Depending on the type of RAM used, the effective and actual frequencies vary. In case of DDR2 and GDDR3 RAM, the effective speeds are twice the actual speeds, since the RAM outputs data twice during every clock cycle.

4.2.20 Mipmapping

This refers to the use of images of differing resolutions depending on the perspective distance between the user and the object in a graphical scene. At close range, the image is used at high resolution to reveal all detail, but as the distance increases, since detail is no longer important, a low-resolution version of the image is used. Doing so saves on resources and doesn't affect the visual quality of the creation. A mipmap is a table that contains the different versions of the various images used in the scene.